

The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, white). On the left side, there's a sidebar with some icons and text. At the bottom, a Windows taskbar is visible with several application icons like Chrome, Photoshop, and others.

C#

Level 1

What is CLR?



C#

Level 1

What is CLR?

🔍 Definition

■ CLR — Common Language Runtime

▨ CLR is the execution engine of the .NET platform.

▨ It provides the environment and services required to run and manage .NET applications.

▨ Main responsibilities include:

- ✓ Executing .NET code
- ✓ JIT compilation
- ✓ Memory management
- ✓ Garbage collection
- ✓ Exception handling
- ✓ Type safety
- ✓ Thread management

➡ CLR = The runtime engine that runs and manages .NET applications.

Explanation

When you write a program in *C#*, the computer does not directly execute the *C#* source code.

C# / *F#* / *VB*



Language Compiler



CIL / *IL*



Assembly (.dll / .exe)



CLR



JIT Compiler



Machine Code



CPU

➡ The CLR is the runtime layer between the .NET application and the operating system/hardware.

▒ CLR as a Runtime Manager

▨ The CLR does not simply translate code.

▨ It also provides runtime services that manage how the application executes.

Procedures

1. Write source code.
2. Compile the source code into IL/CIL.
3. Store IL and metadata in an assembly.
4. CLR loads and manages the assembly.
5. JIT converts required IL into native machine code when applicable.
6. CLR provides runtime services while the application executes.

💡 Example

```
int x = 10;  
int y = 20;  
Console.WriteLine(x + y);  
C# Source Code
```



C# Compiler



IL / CIL



Assembly



CLR



JIT



Machine Code



CPU



Output: 30

➡ The CLR manages the runtime execution of the application.

Note

- ◆ CIL, IL, and historically MSIL refer to the intermediate code used in the .NET execution model.
- ◆ CLR is not a programming language.
- ◆ CLR is not the same thing as .NET.

.NET



Platform



Includes runtime + libraries + tools

CLR



Runtime / Execution Engine

⊘ Warning

- ◆ Do not say:

"CLR converts *C#* directly into machine code."

- ◆ A more accurate flow is:

C#



Language Compiler



IL / CIL



CLR / JIT



Machine Code

- ◆ Code still needs a compatible .NET runtime and supported target platform.



Comparison



CLR

- ✓ Runtime execution environment
- ✓ Manages .NET code execution
- ✓ Provides garbage collection
- ✓ Handles exceptions
- ✓ Provides threading and runtime services
- ✓ Contains/uses the JIT compiler



JIT

- ✓ Part of the runtime execution process
- ✓ Converts IL into native machine code
- ✓ Works during application execution
- ✓ Produces code for the target architecture

✦ Summary

- ✓ CLR = Common Language Runtime
- ✓ CLR is the execution engine/runtime environment for .NET applications
- ✓ Source code is first compiled into IL/CIL
- ✓ IL is stored in assemblies such as .dll and .exe
- ✓ CLR manages the execution of .NET code
- ✓ JIT converts IL into native machine code at runtime
- ✓ CLR provides memory management
- ✓ CLR provides Garbage Collection
- ✓ CLR supports exception handling
- ✓ CLR provides type-safety mechanisms
- ✓ CLR supports thread management
- ✓ CLR ≠ JIT

Thank you 😊